

Basic Math Functions

1. Addition: Addition is the process of combining two or more numbers. Example: $5 + 3 = 8$
2. Subtraction: Subtraction is taking one number away from another. Example: $10 - 4 = 6$
3. Multiplication: Multiplication is repeated addition. Example: $4 \times 3 = 12$
4. Division: Division splits a number into equal parts. Example: $12 \div 3 = 4$
5. Exponents: Exponents represent repeated multiplication. Example: $2^3 = 2 \times 2 \times 2 = 8$
6. Square Root: The square root of a number is a value that, when multiplied by itself, gives the original number. Example: $\sqrt{16} = 4$

Advanced Basic Functions

1. Absolute Value: The absolute value of a number is its distance from zero. Example: $|-5| = 5$
2. Factorial: Factorial of a number is the product of all positive integers less than or equal to that number. Example: $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$
3. Logarithm: A logarithm is the inverse of exponentiation. Example: $\log_{10}(100) = 2$
4. Trigonometric Functions: - $\sin(x)$, $\cos(x)$, $\tan(x)$ Example: $\sin(90^\circ) = 1$
5. Linear Function: A function of the form $f(x) = mx + b$ Example: $f(x) = 2x + 3$
6. Quadratic Function: A function of the form $f(x) = ax^2 + bx + c$ Example: $f(x) = x^2 + 2x + 1$